Code No: 126AQ **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, March – 2024
INFORMATION SECURITY
(Computer science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART - A

(25 Marks)

1.a)	Define security mechanism.	[2]
b)	What do you mean by Cipher text?	[3]
c)	Specify the applications of the public key cryptosystem.	[2]
d)	What is the primitive root of a number?	[3]
e)	Define the term message digest.	[2]
f)	How digital signatures differ from authentication protocols?	[3]
g)	List MIME content types.	[2]
h)	Why does PGP generate a signature before applying compression?	[3]
i)	Identify the three classes of Intruders.	[2]
j)	What are the limitations of firewalls?	[3]

PART - B

(50 Marks)

2.a) Describe the differences between steganography and cryptography with example in details.
b) What is meant by security service? Explain various security services. [5+5]

OR

3.a) Explain Hill cipher with an example.
b) Specify and explain in details the four categories of security threats. [5+5]

4.a) Explain RSA algorithm in detail. Illustrate with an example.
b) Briefly explain the characteristics and operations of RC4 Encryption algorithm. [5+5]

OR

5.a) Explain the principles of public key cryptography.
b) Explain the key management of public key encryption in detail. [5+5]

6.a) Explain the different approaches to message authentication.
b) Illustrate the steps involved in Signature generation and Verification functions of DSS. [5+5]

OR

7.a) Write and explain Client/ Server Authentication Exchange service in Kerberos version 4.

b) What is Public Key certificate? Explain its usage with X.509 certificates. [5+5]

8.a) Write note on PGP session keys, public/private key rings and passphrase keys.

b) What is Radix 64 format? What is its use in PGP? [5+5]

OR

9.a) What is the need to combine Security Associations? Explain basic combinations of Security Associations.

b) Explain the scope of ESP encryption and authentication in tunnel mode. [5+5]

10.a) Explain different Web security requirements.

b) Explain the steps involved in performing Secure Inter-branch Payment Transactions. [5+5]

OR

11.a) Explain Intrusion Detection System (IDS) in detail with suitable diagrams.

b) Illustrate the various types of firewalls with neat diagrams. [5+5]

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